INFO 4220 – Networks II: Market Design

Spring 2022 – Problem Set 2

Solution + rubric

Problem 1 (12 points)

1.1 Determine if the supply and/or the demand curves will shift, and if so, in which direction, in each of the following scenarios. Please explain your rationale clearly and provide any assumption you made if needed.

a. Consider a market composed by insurance companies that sell insurance against natural disasters, and individuals that purchase such insurance. After a natural disaster in a distant country, the Government has decided to make insurance mandatory, establishing that each homeowner must purchase natural disaster insurance. What do you expect will happen in the short term (the short-term assumption means that no new insurance companies have been created, and citizens have not moved away to a different country in response to the new requirements)? (1 point)

The demand curve will move outward, so at the same point (for example at a given price), there will be greater demand (demand at that price will be higher) (1 point).
Point is awarded either with a good written explanation and/or graphs.

b. The government has decided to eliminate taxes applied to the sale of books in an effort to increase literacy. Would this be different than the government providing consumers with a voucher to buy books (this is equivalent to giving people money to buy books)? Explain in terms of what happens to the curves or supply and demand in each case (2 points)

Eliminating taxes affects suppliers, moving the **supply curve** outwards. If a buyer paid P before the elimination of the tax, the seller only keep P – tax. After the tax eliminated, the seller keeps P. Thus, sellers will be willing to sell more quantity at the same price. (1 point).

The voucher moves the **demand curve** outwards since it makes the consumers feel the price is lower (they pay part of the price with the voucher). Thus, consumers will buy more at the same price with the voucher. (1 point).

Award the points only if they note the two cases move a different curve. Explanations can be done written and with graphs.

1.2 You arrive at a dance party with a friend (you are the last 2 guests to arrive) and find there are two potential dance partners available to dance with. For some unknown reason, who dances with whom is determined using the Deferred Acceptance Algorithm with the last 2 guests to arrive proposing. Your preferences and friend's preferences are known. However, the preferences of your potential partners are not. Thus, before running the algorithm you must ask them for their preferences. The preferences you obtain are as follows:

A_1 (you)	A_2 (your friend)
P_1	P_2
P_2	P_1

P_1	P_2
A_2	A_1
P_1	A_2

a. Find the matching. Is it stable? Is there any other stable matching that would be preferred by you and your friend? (1 point)

DA
Step 1
A1 ->P1 (rejected)
A2 -> P2
Step 2
A1 -> P2 (A2 rejects P2).
Step 3
A2 -> P1

Matching:
A1 – P2
A2 – P1
(0.5 points)

There is no better matching that is stable (0.5 points).

b. While you're dancing, you hear P3 laughing because P1 faked their preferences that were, in fact, (A2, A1). Would that change the matching from part A? If there is a new pairing, for whom is it optimum? Use this to explain why P1 lied. (1 point)

Yes, the matching would be (A1, P1) and (A2, P2) which is the most preferred to A and least preferred for P. (0.5 points).

P1 lied because DA is not strategyproof for the receiving party, so P1 faked their preferences to get a matching that they liked better. (0.5 points).

c. Now that you know the true preferences of P_1, will you want to change your dance partner, and will that be possible? (Explain your answer based on the presence or absence of blocking pairs assuming that your current dance partners are those from part a, and the preferences are as in part b) (1 point)

It is not possible to switch because you need a blocking pair. However, **(A1, P2) is not a blocking pair and neither is (A2, P1).** (1 point).
Answer must explicitly mention and use the concept of blocking pairs correctly.

1.3 Consider the Medical Matching problem. Assume that each hospital has a capacity of 1 doctor. Additionally, the government has issued a new law that establishes that hospitals cannot express preferences over doctors and must accept whatever match is determined by the medical matching clearinghouse. The way the new matching works is that doctors submit their preferences over hospitals, and then the clearinghouse determines which doctor goes to which hospital. Which algorithm would you use to solve this problem and why? (3 points)

With no preferences and capacity of 1, the problem becomes a 1-1 assignment (1 point) without endowments (1 point).
We can use serial dictatorship to find an efficient assignment (1 point).

1.4 You are discussing with three friends how to divide an assignment with four questions (obviously not this one, as everyone is expected to solve their own problem set). Each of you lists your preferred questions based on a quick read of the problems:

Person 1: A, B, C, D
Person 2: B, C, D, A
Person 3: A, D, C, B
Person 4: D, C, B, A

a. You ask a neutral friend to assign the problems in the way they see fit. Your friend says that they will just ask each of you, in the order P1, P2, P3, and P4, for the problem they want to solve. When is your turn you can ask for any problem that has not been assigned yet. Is this allocation mechanism strategyproof? Prove why or why not. (1 point)

Using serial dictatorship, we get (P1, A), (P2, B), (P3, D), and (P4, C) (0.5 points).

It is strategyproof because, with the order provided, participants have no incentives to fake their preferences since there is no way they can get a better assignment [P1 and P2 has their favorite problems, P3 has 2nd favorite and faking preferences won't get them A, and P4 has its 2nd favorite and faking preferences would not get them D]. (0.5 points).
Note: Saying that the mechanism described is a serial dictatorships and serial dictatorship are not strategyproof is not a valid proof.

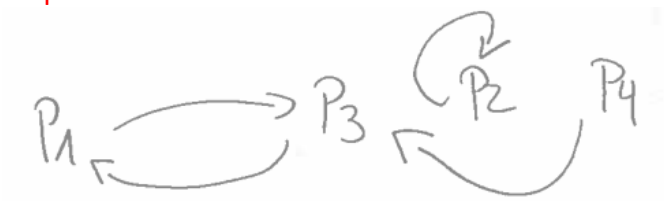
b. After a day working on the problem set, some of you realize that your preferences from the day before were wrong and would like to change problems. Thus, you decide to redistribute the questions. To avoid conflicts, you decide that each person "owns" the problem they had been assigned the day before and are not obligated to switch problems if they don't want to. Considering the new preferences shown below. What

algorithm would you use to re-assign problems? Use the algorithm to find the resulting assignment. (2 points)

Person 1: D, B, C, A
Person 2: B, A, C, D
Person 3: A, D, C, B
Person 4: D, B, C, A

TTC with endowments (0.5 points).

Step 1



We get the following assignments from the 2 cycles:
(P1, D)
(P3, A)
(P2, B)

(1 point).

Step 2:
P4 → P4
(P4, C).
(0.5 points).

Problem 2 (14 points)

2.1 Consider a medical matching problem with four hospitals, h_1, h_2, h_3, and h_4, and five doctors, d_1, d_2, d_3, d_4, and d_5. All hospitals have different capacities for hiring doctors. $h_1$and h_2 can admit up to 1 doctor, h_3 and h_4 can admit up to 2 doctors. The following tables give the preferences of doctors over hospitals and hospitals over doctors:

d_1	d_2	d_3	d_4	d_5
h_4	h_3	h_3	h_3	h_2
h_3	h_2	h_4	h_1	h_1
h_2	h_1	h_1	h_2	h_3
h_1	h_4	h_2	h_4	h_4

h_1	h_2	h_3	h_4
d_2	d_5	d_3	d_1
d_1	d_2	d_4	d_5
d_3	d_3	d_1	d_3
d_4	d_1	d_2	d_2
d_5	d_4	d_5	d_4

a. Find the matching that would result from using the deferred acceptance algorithm with doctors proposing. (2 pts)

Step 1:
Proposals:
d1 → h4
d2 → h3 (d2 rejected)
d3 → h3
d4 → h3
d5 →h2
(1 point)

Step 2:
Proposals:
D2 → h2 (d2 rejected).
(0.5 points).

Step 3
Proposals:
D2 → h1

Result:
(h1, d2), (h2, d5), (h3, d3 & d4), (h4, d1).
(0.5 points).

b. After the matching, doctors sign a 1-year contract. At the end of the year, they have the option to extend their contract for another year (and the hospital cannot refuse to extend their contract). If they don't extend their contract they participate in a new matching in which hospitals and doctor can revise their preferences. You can think of the process as two-step system. In step 1 doctors announce if they will extend their contract. In step 2, a new matching is run between all doctors that did not extend their contract and hospitals with open spots using their revised preferences. All doctors have different expertise in medical fields. d_1 is an expert in dental care, d_2 is an expert in the treatment of children, d_3 is an expert in stomach problems, and d_4 is an expert in neck and nose problems. As the birthrate has decreased, small hospitals (h_1 and h_2) where only one doctor can be hired decide to put their preference of doctors for children to the least from the next contract. Hospital 3 has decided that they did not like working with d_4 and have moved them to the bottom of their preference list. Additionally, d_5 decides not to extend their contract and not to participate in the new matching due to personal problems. Find the new allocations that are made after the end of year 1. Will this matching be stable? (3 points)

d5 quits (0.5 points).
d1, d3, d4 extend contracts because they have their favorite hospital (0.5 point).

d2 participates in the market because they are not matched to their favorite hospital and there are two hospitals with open seats (h1 and h2). (0.5 points).

There are revised preferences:

h_1	h_2	h_3	h_4
d_1	d_5	d_3	d_1
d_3	d_3	d_1	d_5
d_4	d_1	d_2	d_3
d_5	d_4	d_5	d_2
d_2	d_2	d_4	d_4

(0.5 points).

Step 1: d2 -> h3 (d2 rejected because h3 is full).
Step 2: d2 -> h2
(A valid alternative solution is saying that h3 and h4 are not participating because they are full (and cannot reject the doctors they already have). In that case d2 proposes to h2 directly)

Assignment will be (h1, o), (h2, d2), (h3, d3 & d4), and (h4, d1).
(0.5 points).

The assignment is not stable because (d2, h3) are a blocking pair (0.5 points).

c. In general, what is the problem with using matching mechanisms that don't produce stable matchings in the medical match problem? (2 points)

When a matching mechanisms that is not stable is used, it means that the resulting matching could contain blocking pairs, matchings that are not individually rational, or wasteful matchings. If there is a blocking pair, the doctor and hospital in the pair will want to not honor the match they received and match between them instead. A doctor that receives a matching that is not individually rational can refuse to work for the hospital they were assigned to. If the matching is wasteful, some doctors can move to hospitals that they like better and that have open spots to receive them. When all these things happen, the markets (i.e. the clearinghouse doing the matchings) tend to fail in the long term and the matching system is abandoned. (2 points).
How to assign points: Key aspect is to say the market will likely collapse (1.5 points). The other 0.5 points assign them if there is a reasonable explanation (i.e. that doctors/hospitals will have incentives to not respect the matching they received)

2.2 Consider the following medical match with couples. There is one couple (Alice, Bob), and one single doctor (Carol). There are two hospitals that can hire 1 doctor each. The preferences of doctors and hospitals are as follows:

(Alice, Bob)	Carol
(h_1, h_2)	h_1
(Alice, Bob)	h_2

h_1	h_2
Alice	Carol
Carol	Bob

Can you find a stable matching? Find a stable matching or demonstrate it is impossible to have a stable matching in this problem. (7 points)

To prove that there is no stable matching, we can consider all the possible ways of matching the doctors/hospitals:

- One doctor of the couple to one hospital and Carol to the other is not stable because (Alice, Bob) will block (1.5 points).
- (Alice, H1) (Bob, H2) → (Carol, H2) will block (1.5 point).
- (Carol, H1), (o, H2) → (Alice, Bob) and (h1, h2) Will block (1.5 point).
- (0, h1), (Carol, h2) → (Carol, h1) will block (1.5 point).

Conclude that it is impossible to have a stable matching (1 points). Or if they made a mistake along the way, 1 point if they tried to prove this by going over all potential matchings.

Problem 3 (12 points)

Greek Life is an example of an organization at Cornell where membership seniority determines the priority in selecting rooms within the communal housing. A new Fraternity, Nu Epsilon Tau (NET), was chartered at Cornell last year. Usually, after new members join in the fall, they gather the organization together to select the rooms that each member will live in the following year. Let's say there are three current members, i_1, i_2, and i_3 returning from those that lived in the house last year, living in rooms r_4, r_2, and r_3, respectively. There are two new members this year, i_4 and i_5. In addition to the three currently occupied rooms, there are three additional rooms, r_1, r_5, and r_6, whose current occupants are graduating. These rooms are bigger and have ensuite bathrooms, but the rent is also significantly higher.

Suppose the five students' preferences over the six rooms is as follows:

i_1	i_2	i_3	i_4	i_5
r_2	r_2	r_3	r_3	r_5
r_5	r_1	r_5	r_2	r_4
r_4	r_3	r_4	r_1	r_3
r_3	r_5	r_2	r_5	r_1
r_1	r_4	r_1	r_4	r_2

a. The younger members want to try modernizing the way rooms are assigned. Instead of basing it on seniority order as usual, they will draw lots to determine the order in which each member selects their room. However, any one of the older members i_1, i_2, i_3 can veto the outcome, in which case all older members stay in their current rooms and the younger members must select a room among those available. Suppose the result of drawing lots is: i_1, i_2, i_4, i_5, i_3. What would be the room assignment without considering the potential veto by senior members? Would any of them be inclined to veto this method and keep their current room instead? (2 points)

Serial Dictatorship
Rooms chosen by dictators:
I1 → r2
I2 → r1
I3 → r4
I4 → r3
I5 → r5
(1 point)
I2 and i3 will veto because they end up worse (1 point).

b. Is this assignment (the one from part a) efficient? If so, explain why, and if not, provide an example to show this. (1 point)

It is efficient because there is no way to make someone better without worsening someone (1 point).

c. When a senior member vetoes the assignment, all senior members keep their current room, and the new members must choose one of the available rooms using the same ordering as in (a). Find the assignment that would result if a senior member vetoes the assignment. Is this assignment efficient? Explain why or why not. (3 points)

i1, i2, i3 keep their current room:
I1 → r4
I2 → r2
I3 → r3
i4, i5 choose a room from the rooms remaining:
I4 → r1
I5 → r5
(1.5 point)
It is efficient because you cannot make someone better without making someone else worse (1.5 points).

d. A senior member (i_1) who is currently enrolled in Networks 2 is arguing that both methods are unfair to senior members. Because he currently has a room he doesn't like much, while the other two senior members have their favorite rooms. Thus, it is very

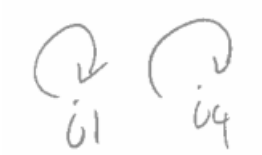
likely that one of them will veto the assignment and, when that happens, he will be stuck in one of his least preferred rooms. And even if no one vetoes the assignment, he is not getting any preference given his seniority. He proposes the following modification. After someone vetoes the assignment and rooms are assigned as calculated in part c, a new step is added in which the TTC algorithm is used to improve the assignments. In that way, senior members can trade their existing room for a better one if possible. Find the assignment that would result from this method. Is it any better for i_1? Explain why or why not. (2 points)

Step 1:



3 Cycles:
I2 → i2 (keeps r2).
I3 → i3 (keeps r3)
I5 → i5 (keeps r5)
(1 point).

Step 2:



2 Cycles:
I1 → i1 (keeps r4)
I4 → i4 (keeps r1)
(0.5 points).

The result is the same as in c), so i1 is not better (0.5 points).

e. If you had to suggest an algorithm to use in this situation, what would you use and why? Implement it and find the assignment that would result. (4 points)

YRMH-IGYT or TTC with mixed endowments which would be strategyproof, efficient, and individually rational (1 point).

Option 1 YRMH-IGYT (2 points if well implemented)

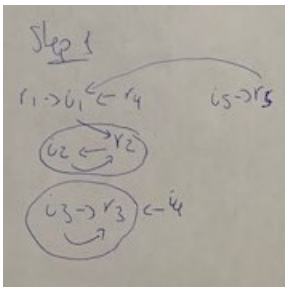
(i1, r4), (i2, r2), (i3,r3) order i1, i2, i4, i5, i3

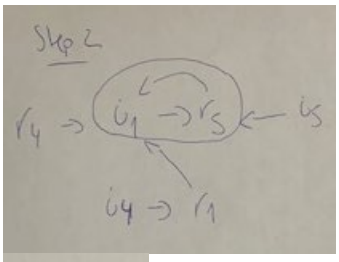
Step 1
i1 requests r2. There is a conflict. We re-order: i2, i1, i4, i5, i3

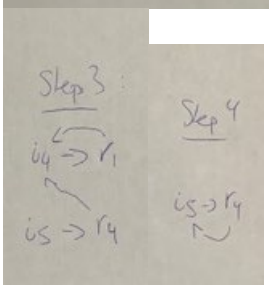
Step 2
i2 requests r2
i1 requests r5
i4 requests r3 . There is a conflict. We re-order: i3, i4, i5 and resume from i3

Step 3
i3 requests r3
i4 requests r1
i5 requests r4

Option 2 TTC with mixed endowments (2 points if well implemented).







Result (1 point)
(i1, r5)
(i2, r2)
(i3, r3)
(i4,r1)
(i5,r4)

Problem 4 (12 points)

A well-established company recently opened three new factories in three bustling cities in the United States - New York, NY; Los Angeles, CA; and Dallas, TX. In order to ensure seamless operations and maximize productivity, the company has decided to appoint experienced individuals, Alice, Bob, and Amy, as the factory managers for these new facilities. Each of the managers has their preferences over which of these cities they want to work at:

Alice	**Bob**	**Amy**
Dallas	New York	New York
Los Angeles	Los Angeles	Dallas
New York	Dallas	Los Angeles

a. After carefully considering the location preferences of Alice, Bob, and Amy, the company has noted that both Bob and Amy have a strong tendency to work in New York. Given this information, the company has decided to use the probabilistic serial algorithm (eating algorithm) to determine the assignment of managers to the three factories. Find the resulting random assignment. (3 points)

Time 0: Alice-Dallas, Bob & Amy – NYC.
Time ½: NYC fully eaten, Dallas ½ Eaten. Bob starts eating LA, Amy starts eating Dallas, Alice continues eating Dallas.
(1 point)
Time 3/4: Dallas is fully eaten and ¼ LA is eaten. All start eating LA.
Time 1: LA is eaten.
(1 point)

Result

	Dallas	NYC	LA
Alice	¾	0	¼
Bob	0	½	½
Amy	¼	½	¼

(1 point)

b. Suppose instead that the managers are assigned to factories via a random serial dictatorship. Find the resulting random assignment. (3 points)

We do all the potential orders (6) (2 points)

Alice, Bob, Amy: (Alice, Dallas) (Bob, NYC) (Amy, LA)
Alice, Amy, Bob: (Alice, Dallas) (Amy NYC) (Bob, LA)

Bob, Alice, Amy: (Bob, NYC) (Alice, Dallas) (Alice, LA)
Bob, Amy, Alice: (Bob, NYC) (Amy, Dallas) (Alice, LA)
Amy, Bob, Alice: (Amy, NYC) (Bob, LA) (Alice, Dallas)
Amy, Alice, Bob: (Amy, NYC) (Alice, Dallas) (Bob, LA)

Result

	Dallas	NYC	LA
Alice	5/6	0	1/6
Bob	0	½	½
Amy	1/6	½	1/3

(1 point)

c. The manager in charge cannot decide which method to choose. He has decided that Alice, Bob, and Amy can vote for the method they prefer, and he will use the method that gets the most votes. Which method will be chosen? (Hint: Think of stochastic dominance) (3 points)

Alice	PS	RSD
Dallas	3/4	5/6
Dallas, LA	1	1

Alice prefers RSD.
(1 point)

Amy	PS	RSD
NYC	1/2	½
NYC, Dallas	3/4	2/3

Amy prefers PS
(1 point)

Bob	PS	RSD
NYC	½	½
NYC, LA	1	1

Bob is indifferent
(0.5 points).

Voting will not yield a result based on their preferences. Bob will decide whether he is influenced by external factors or randomly because he is indifferent (0.5 points).

d. After a method is decided in part (c), the same manager in charge is still very confused, because they need to assign someone to a city, and they don't know what to do with the matrix of probabilities the method is giving them. Explain a way in which the

manager can select a deterministic assignment in a way that is consistent with the result in part (c). Is it possible that your method results in an assignment that is not efficient? (3 points)

The random assignment (either of the two) has to be decomposed into a deterministic assignment and then a lottery needs to be designed.

Option 1 (PS)

$$\begin{pmatrix} 3/4 & 0 & 1/4 \\ 0 & 1/2 & 1/2 \\ 1/4 & 1/2 & 1/4 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} + \frac{1}{4}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \frac{1}{4}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

(2 points)

We need a random number generator between 1 and 4. Depending on the number that comes out, we choose an assignment in the following way: 1 & 2 are the first matrix, 3 the second and 4 the last (or any other equivalent method). (1 point).

Option 2 (RSD)

$$\begin{pmatrix} 5/6 & 0 & 1/6 \\ 0 & 1/2 & 1/2 \\ 1/6 & 1/2 & 1/2 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} + \frac{1}{3}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \frac{1}{6}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

(2 points)

We need a dice (regular 6 sides). We throw the dice and choose an assignment based on the number that comes out: 1-3 will choose the first matrix, 4-5 the second, and 6 the last (or any other equivalent method). (1 point).